

Project Initialization & Planning Phase

Define Problem Statements

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Project Name	Smart Lender: AI-Based Loan Eligibility Prediction System
Maximum Marks	3 Marks

Project Problem Statement

- Traditional loan application processing in financial institutions relies heavily on manual evaluation, leading to high processing times, high operational overhead, and potential human bias.
- Existing banking systems often struggle to provide instant eligibility feedback and transparent risk assessments due to rigid scoring rules that cannot scale.
- This project aims to develop a modular, machine learning-driven loan eligibility prediction system ('Smart Lender') that cleans, processes, and evaluates historical applicant data to generate instant predictions.
- Integrating multiple classification algorithms (Decision Tree, Random Forest, KNN, XGBoost) and deploying them through a Flask application provides credit risk analysts with automatic pre-screening, batch prediction, and an interactive analytical trends dashboard.

Customer Problem Statement

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
Financial Institution / Business Perspective					
PS-1	A Credit Risk Analyst / Loan Officer	Evaluate high volumes of loan applications quickly and accurately.	Manual processing of applicant data is slow, prone to human error, and inconsistent.	Assessing eligibility requires examining numerous complex financial variables (income, credit score, debt,	Stressed by severe backlogs of pending applications and anxious about potential high-risk default approvals.

				assets) simultaneously.	
Applicant / Customer Perspective					
PS-2	A Loan Applicant (Retail / Business)	Secure a loan decision quickly to plan business or personal milestones.	Traditional manual banking pipelines take days or weeks with very low process transparency.	Evaluation depends on opaque, manual validation processes of multi-tier criteria that lack instant status feedback.	Anxious and frustrated about the delay, making financial and personal planning extremely difficult.